

1. An Interface in Java programming language is defined as an abstract type used to specify the behavior of a class.

An interface in Java is a blueprint of a class. It has static constants and abstract methods.

2. Syntax:

Interfaces Abstract Class

1. Can't provide the implementation of abstract class. 1. Can provide the implementation of interface.

2. Members of java interface are public . 2. Members of java interface can be public, protected etc.

3. A class can implement multiple interface. 3. A class can inherit only one Abstract class.
(Multiple Inheritance) (No Multiple Inheritance)

4. They can only have abstract methods. 4. They can have both abstract as well as non-abstract method.

5. Interface has only static and final variables. 5. Can have final, non-final, static and non-static variables.

6. Can extend another Java interface only. 6. Can extend another Java class and implement multiple Java interfaces.

7. Can be implemented using keyword "implements". 7. Can be extended using keyword "extends".

8. Example: 8. Example:

```
public interface Drawable {      public abstract class Shape {  
    void draw();      public abstract void draw();  
}      }
```

3. Every other rule is same for Interfaces as in Inheritance.

For example, If you "Engine" Interface and "Car" Class then you can have:

Car car = new Car();

as well as Engine car = new Car();

All the rules applied will be same as before.

4. Both in Inheritance as well as Interfaces, overridden methods should be with same access modifier as in parent class

(or interface) or it should be better.

For example if the method which need to be overridden is protected, then in subclass it should be protected or better

such as public.